NIELIT Virtual Academy

INTERSHIP PROSPECTUS

Name of the Internship: Online Internship in Embedded C

Internship Code: IN11

Mode of Conduction: Online

Starting Date: 2nd June 2025

Last date of registration: 1st June 2025

Duration: 6 Weeks

Objective of the Course

- Provide basic theoretical concepts embedded system concepts and Embedded 'C'
- Provide a comprehensive understanding of Memory management and Pointers,
- Provide a deep understanding of the architecture of ARM Cortex processor
- Provide hands-on for highly deterministic real-time applications

Outcome of the Course

- Provide the participants with in-depth knowledge of Embedded C programming in an ARM microcontroller environment.

Course Fee: **Rs: 1000/-(inclusive of GST)**

Eligibility: **Pursuing an Undergraduate level course or above**

Methodology

- ✓ Teaching Mode: Self-Pace
- ✓ Access from anywhere anytime
- ✓ Content Access through e-learning portal
- ✓ Doubt Removal Session
- ✓ Covers both Theory & Practical
- ✓ Certification: On completion of the Mini Project

Registration Link: http://nva.nielit.gov.in

Contact Details:

- ✓ Course coordinator Name: Mr. Raghvendran S
- ✓ Email: trng.chennai@nielit.gov.in; contact.nva@nielit.gov.in
- ✓ Mobile number: **7598730125**

Course Structure:

Module No	Module Title
1	Embedded Concepts
2	C and Embedded C
3	Introduction to ARM Cortex
4	Cortex M4 Microcontrollers & Peripherals
5	Project

Syllabus:

Detailed Syllabus

Module 1: Embedded Concepts

- ✓ Introduction to embedded systems
- ✓ Application Areas
- ✓ Categories of embedded systems
- ✓ Overview of embedded system architecture
- ✓ Specialties of embedded systems
- ✓ Recent trends in embedded systems
- ✓ Hardware architecture
- ✓ Software architecture
- ✓ Application Software
- ✓ Communication Software
- ✓ Development and debugging Tools

Module 2: 'C' and Embedded-C

- ✓ Introduction to 'C' programming
- ✓ Storage Classes
- ✓ Data Types
- ✓ Controlling program flow
- ✓ Arrays
- ✓ Functions
- ✓ Memory Management
- ✓ Pointers
- ✓ Arrays and Pointers
- ✓ Structures and Unions
- ✓ Data Structures
- ✓ Linked List
- ✓ Stacks, Queues
- ✓ Conditional Compilation
- ✓ Pre-processor directives
- ✓ Bitwise operations
- ✓ Typecasting

Module 3: Introduction to ARM Cortex

- ✓ Architecture Introduction to 32-bit Processors
- ✓ The ARM Architecture
- ✓ Overview of ARM
- ✓ Overview of Cortex Architecture
- ✓ Cortex M4 Register Set and Modes
- ✓ Cortex M4 Processor Core
- ✓ ARM Cortex M4 Development Environment

Module 4: Cortex M4 Microcontrollers & Peripherals

- ✓ Cortex M4 based controller architecture
- ✓ Memory mapping, Cortex M4 Peripherals – RCC
- ✓ GPIO
- ✓ Timer, System timer
- ✓ UARTs
- ✓ Cortex M4 interrupt handling – NVIC
- ✓ Application development with Cortex M4 controllers using standard peripheral libraries

Module 5. Project

- ❖ Capstone project
- ❖ Project Report Submission